

Glint and Glare Assessment

Millbrook Solar & Storage – Lumen Reflect Consulting, report LRC-2026-044 Rev 2

Sofie Lindqvist · 29 May 2026 · MB-DES-RPT-0021

Summary

Lumen Reflect Consulting was commissioned by Weyland Energy to assess the potential for glint and glare from the proposed Millbrook Solar & Storage scheme. We modelled 34 residential receptors, 5.5 km of road, the branch railway and the approach paths of Millbrook Airfield, using the submitted layout (Rev C) with single-axis trackers ($\pm 55^\circ$ rotation, backtracking enabled, 2.9 m maximum height).

Result: with the proposed 3 m hedgerow screen along the Mill Lane frontage, no receptor experiences more than a *low* residual effect. No mitigation is needed for road users, rail or aviation.

1. Methodology

Reflections were modelled with a geometric ray-tracing model at one-minute intervals for a full year, for each receptor at 1.5 m (ground floor) and 4.5 m (first floor) above ground. Bare-earth terrain was used first; existing hedgerows and buildings were then added from a drone survey of 14 March 2026. Effects are classified as None, Low, Moderate or High, considering duration (minutes per day, days per year), the time of day and whether the sun is in the same field of view.

2. Receptors and results

ID	Receptor	Type	Distance to panels (m)	Geometric potential	Unmitigated	Mitigated
R01	Ashcombe Farmhouse	Dwelling	380	No	None	None
R02	Ashcombe Barn	Dwelling	420	No	None	None
R03	1 Mill Lane	Dwelling	210	Yes	Low	None
R04	Mill Lane Cottages 1	Dwelling	95	Yes	Moderate	None
R05	Mill Lane Cottages 2	Dwelling	90	Yes	Moderate	None
R06	Mill Lane Cottages 3	Dwelling	105	Yes	Moderate	None
R07	The Old Mill	Dwelling	260	Yes	Low	None
R08	Hollins Farmhouse	Dwelling	540	Yes	Low	None
R09	Brook House	Dwelling	610	No	None	None
R10	Beacon View	Dwelling	880	No	None	None
Road 1	Mill Lane (2.1 km section)	Road	adjacent	No	None	None
Road 2	B4127 (3.4	Road	adjacent	Yes	Low	None

	km section)					
Rail	Branch line north of the Site	Railway	1450	No	None	None
Avia 1	Millbrook Airfield (microlights), 7.2 km	Aviation	7200	No	None	None

Extract of the 34 residential receptors (all receptors are listed in Appendix B)

3. Discussion

Mill Lane Cottages (R04–R06) sit on rising ground 90–105 m south of field 3. In the early morning between late April and mid-August, reflections could be seen from first-floor windows for up to 24 minutes per day when the trackers are close to horizontal at the start of the backtracking period. The existing roadside hedge is 1.4–1.8 m high and gappy. Without mitigation the effect is *moderate*.

A new native hedgerow along the northern side of Mill Lane, maintained at a minimum of 3 m (planted at 1.8 m with a temporary 3 m close-boarded screen until established), removes every modelled reflection path to R04–R06. We recommend that the hedgerow height is secured by planning condition.

For the B4127, reflections were possible at two short sections but always within 30° of the direction of the sun, i.e. the driver is already looking into the sun. The effect is *low* and no mitigation is required.

4. Recommendations

1. Plant a 3 m native hedgerow along the Mill Lane frontage of fields 2 and 3, with a temporary screen during establishment.
2. Keep backtracking enabled; do not use a fixed stow angle above 10° during early morning hours.
3. Re-run the model if the tracker supplier or tracker rotation limits change after EPC award.

IMAGE PLACEHOLDER

millbrook-site-mill-lane

Narrow single-track country lane with grass verges and a gappy low hedgerow, three brick cottages on rising ground, a farm tractor in the distance, early spring, overcast.

Mill Lane frontage of field 3 looking north-west, 14 March 2026